MegaRAC Development Kit for the ASPEED AST2600 BMC

The MegaRAC Development Kit for the AST2600 BMC features hardware, tutorials, and access to our MegaRAC Community Edition™ BMC Firmware based on The Linux Foundation® OpenBMC™ firmware. This kit is designed to help developers and engineers gain experience in sensor porting, management technologies such as Serial over LAN (SOL) and UART, and virtual media (vMedia).

The kit allows customers to begin immediate development and training projects using MegaRAC Community Edition BMC firmware with the AST2600 BMC from ASPEED Technology - to deliver projects faster, with experience and confidence.

MegaRAC Community Edition™ Firmware

MegaRAC Community Edition BMC firmware is a hardened, production version based on The Linux Foundation® OpenBMC™ firmware. MegaRAC Community Edition can be enhanced with proprietary AMI IP packages based on the proven MegaRAC SP-X Service Processor firmware stack, trusted for years by leading OEMs and ODMs for its robust, secure remote server management. MegaRAC Community Edition features an intuitive, extensible open-architecture development framework.

Bundled Hardware

- ASPEED AST2600 Evaluation Board (EVB)
- AMI Series 997 Sensor Board
- Host System
- Fan

Included Tutorials

- MegaRAC Community Edition BMC Firmware
- AMI DevNet® Installation and Environment Preparation
- Sample BMC firmware development projects to help gain development experience

Key to its flexibility is a robust "common core" concept that delivers a highly managed source base with high ROI, and a complete service processor solution for wide product deployments. MegaRAC Community Edition includes dedicated support, code patches and security advisories that all AMI customers trust and expect.

MegaRAC Community Edition encompasses a family of blended firmware / software, tools, utilities and services that allow OEMs, ODMs, and CSPs to select the perfect blend of modularity, support and services to meet specific project timelines, budgets and objectives.

MegaRAC FW/SW Solutions	MegaRAC Services
Server Baseboard Management Controller (BMC)	Fee-based services for complete development and delivery of the product
Enclosure (Chassis or Rack Manager) BMC	Turnkey and Engineering (NRE) Services, including BMC feature design and customization
Essential licensable IP modules to provide complete manageability	Hardware Platform Porting
Full-featured AMI development environment, which offers distinct advantages and benefits for FW enhancement	Quality Assurance (QA) & Security Analysis Services

Materials Needed

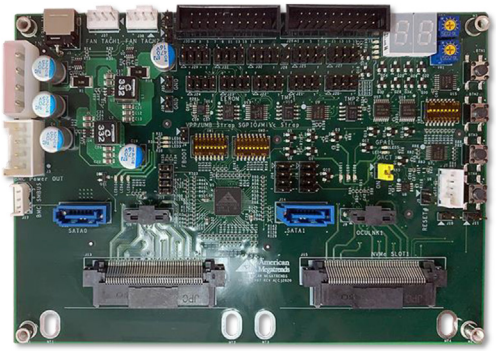
- USB Driver (SD, USB flash for virtual media)
- Ethernet, Serial and/or Display Cable(s)

For more information please visit: ami.com/megarac-development-kit-for-the-ast2600-bmc/

MegaRAC firmware supports a number of Customer Reference Board (CRB) designs, with new platform support being added on a periodic basis.

AMI Series 997 Sensor Board

The Series 997 Sensor Board is designed to mimic the sensors that are typically found on data center server platforms. This allows ODM and OEM customers to begin immediate development using MegaRAC Community Edition BMC firmware with the AST2600 BMC. Firmware engineers gain hands-on experience with sensor porting, management via Serial Over LAN, management via UART and with vMedia.



- 2x Current/ voltage sensors (INA219A) for 12V and 5V DC power monitoring
- 2x drive facing connectors for NVMe/SAS/ SATA (SFF-8639 U.2 Slot; U.3 compatible)
- 2x host facing connectors for U.2/U.3 NVMe (x4 OCulink)
- 2x host facing connectors for SATA/SAS (std 7-pin SATA)
- 1x Backplane Controller (AMI MG9100) UBM/ VPP/SGPIO host protocol for drive, LED
- 1x NVMe I²C management interface (accessible via pin header)
- 1x 2-channel I²C MUX with Interrupt (PCA9543A)
- 2x temperature sensors (TM75AID)
- 1x 16K bit EEPROM (24LC16B)

AST2600 Evaluation Board

Featuring ASPEED AST2600 Server Management Processor

For more information please visit: ami.com/megarac-development-kit-for-the-ast2600-bmc/

- 1x 2-to-1 I²C master de-multiplexer with an arbiter function (PCA9641)
- 1x I²C repeater (PCA9617)
- 1x Parallel-in/serial-out (74LV165A) with 8
- 2x Serial-in/ parallel-out (74LV595A) with 7 segment LED display
- 2x potentio meters for +12V, +5V voltage monitoring (to BMC ADC input)
- 1x 2-channel +12V DC FAN control (PWM)/ monitoring (TACHO)
- Signals are available via pin headers for composing desired I²C topologies
- 4 GPIO controlled LEDs
- 4 digital inputs via 4 tactile switches

For more information please visit: ami.com/megarac-development-kit-for-the-ast2600-bmc/

The Linux Foundation and OpenBMC™ are a marks of LF Projects, LLC in the US and other countries.

©2025 AMI. All rights reserved. Product specifications are subject to change without notice. Products mentioned herein may be trademarks or registered trademarks of their respective companies. No warranties are made, either expressed or implied, with regard to the contents of this work, its merchantability or fitness for a particular use. This publication contains proprietary information and is protected by copyright. AMI reserves the right to update, change and/or modify this product at any time.